THE JOIN OF TWO NUCLEI AS A LAX MODALITY: AXIOMATIZATION, FINITE MODELS, AND INEXPRESSIBILITY

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Abstract. Propositional lax logic is the intuitionistic modal logic of a nucleus on a Heyting algebra, equivalently of a Lawvere–Tierney topology on a topos. Two nuclei have a join in the lattice of nuclei, whose fixed points are the common fixed points; on a finite algebra it is a finite iterate of the composite, on a site it is the join of the two Grothendieck topologies. We axiomatize the trimodal logic of two lax modalities together with their join by adding to the lax axioms the inclusions $\bigcirc_1\varphi \to \bigcirc\varphi$, $\bigcirc_2\varphi \to \bigcirc\varphi$ and one induction scheme, $(\bigcirc_1\psi \to \psi) \to (\bigcirc_2\psi \to \psi) \to (\varphi \to \psi) \to (\bigcirc\varphi \to \psi)$, and prove algebraic completeness. On a finite poset P the nuclei of $\mathrm{Up}(P)$ are the relativized necessities $\bigcirc_C$ over subsets $C \subseteq P$, the join of $\bigcirc_C$ and $\bigcirc_{C'}$ is $\bigcirc_{C\cap C'}$, and the join equals the $|P|$-fold iterate of the composite, which gives the finite-site axiom. The logic has the finite model property, by a selective filtration of the prime-filter model in which each nucleus is represented by its closed prime filters and the induction scheme forces the points isolated from above to be closed under the join; hence it is decidable and complete for complete Heyting algebras and for Grothendieck toposes. The join is nevertheless not expressible in the bimodal lax logic: on the chain $\omega + 2$ with the two nuclei stepping from even to odd and from odd to even ordinals, every closed term of the two-nucleus language takes a value in $\mathbb{N} \cup \{\omega + 1\}$, while the join of the two nuclei sends $\bot$ to ω. The induction scheme is the lax analogue of the induction axiom of common knowledge, with the join playing the role of the common-knowledge operator and the iterates that of "everybody knows" to depth n.

1. Introduction

A *nucleus* on a Heyting algebra H is a map $j\colon H \to H$ with $a \le ja$, $jja = ja$ and $j(a \wedge b) = ja \wedge jb$. Nuclei are the Lawvere–Tierney topologies of topos theory restricted to the algebra of subobjects of 1, the "geometric modalities" of Goldblatt [7], and the algebraic semantics of the propositional lax logic **PLL** of Fairtlough and Mendler [6]: intuitionistic propositional

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logic with a modality $\bigcirc$ satisfying $\varphi \to \bigcirc\varphi$, $\bigcirc\bigcirc\varphi \to \bigcirc\varphi$ and $\bigcirc\varphi \wedge \bigcirc\psi \to \bigcirc(\varphi \wedge \psi)$.

The nuclei of a complete Heyting algebra form a complete lattice under the pointwise order; meets are pointwise, and the join $j_1 \vee j_2$ is the nucleus whose fixed points are the common fixed points of j_1 and j_2 [10, II.2]. The join is harder to compute: Macnab [12] obtains it as a transfinite iterate of the composite $j_1 j_2$, and on a finite algebra it is a finite iterate. For Grothendieck topologies on a site the join of nuclei is the join of topologies, so a statement about $j_1 \vee j_2$ is a statement about "locally for $J \vee J'$".

This paper treats the join as a modality. Let $\mathbf{PLL}_2$ be the bimodal lax logic with two lax modalities $\bigcirc_1, \bigcirc_2$, and let **LJ** (Definition 3.1) be the trimodal logic obtained by adding a third lax modality $\bigcirc$ with the axioms $\bigcirc_1\varphi \to \bigcirc\varphi$, $\bigcirc_2\varphi \to \bigcirc\varphi$ and the *induction scheme*

$$\text{(JL)} \qquad (\bigcirc_1\psi \to \psi) \to (\bigcirc_2\psi \to \psi) \to (\varphi \to \psi) \to (\bigcirc\varphi \to \psi).$$

Theorem 1.1. (a) **LJ** *is algebraizable, its equivalent algebraic semantics being the Heyting algebras with nuclei j_1, j_2, j in which ja is the least common fixed point of j_1 and j_2 above a, for every a; on complete and on finite Heyting algebras this says that j is the join of j_1 and j_2 in the lattice of nuclei, so every complete Heyting algebra with two nuclei and their join, in particular every Grothendieck topos with two Lawvere–Tierney topologies and their join, is such an algebra (Theorem 3.3).*

(b) *On a finite poset P the nuclei of $\mathrm{Up}(P)$ are the relativized necessities $\bigcirc_C\varphi =$ "φ at every point of C above the current one", $C \subseteq P$; the join of $\bigcirc_C$ and $\bigcirc_{C'}$ is $\bigcirc_{C\cap C'}$ and equals $(\bigcirc_C\bigcirc_{C'})^{|P|}$ (Theorem 4.1). Hence* **LJ** *together with the scheme $(\bigcirc_1\bigcirc_2)^{N+1}\varphi \to (\bigcirc_1\bigcirc_2)^N\varphi$ proves $\bigcirc\varphi \leftrightarrow (\bigcirc_1\bigcirc_2)^N\varphi$, and the scheme holds on every poset with at most N points.*

(c) *The join is not expressible in $\mathbf{PLL}_2$: there is no formula χ in $\bigcirc_1, \bigcirc_2$ such that $\bigcirc\bot \leftrightarrow \chi$ holds in all* **LJ***-algebras, and a fortiori no $\chi(p)$ with $\bigcirc p \leftrightarrow \chi(p)$ (Theorem 5.1).*

(d) **LJ** *has the finite model property: every non-theorem is refuted on a finite poset with two subsets. Hence* **LJ** *is decidable and complete for complete Heyting algebras with two nuclei and their join, in particular for Grothendieck toposes with two Lawvere–Tierney topologies and their join (Theorem 6.4, Corollary 6.5).*

The induction scheme is the intuitionistic, inflationary counterpart of the induction axiom of the logic of common knowledge [9]: common knowledge $C\varphi$ is the greatest fixed point of $X \mapsto E(\varphi \wedge X)$, where E is "everybody knows", and is not a finite iterate $E^n\varphi$; the join $\bigcirc\varphi$ is the least fixed point of $X \mapsto \varphi \vee \bigcirc_1 X \vee \bigcirc_2 X$ (Lemma 2.2) and is not a finite iterate $(\bigcirc_1\bigcirc_2)^n\varphi$.

What is known. Lax logic and its algebraic, Kripke and cover semantics are in [6, 7, 8]. That the join of nuclei on a frame has as fixed points the

common fixed points is classical [10, II.2]; Macnab [12] computes the join by transfinite iteration of the composite, and Escardó [5] computes joins of arbitrary families of nuclei as least fixed points of inflationary operators on prenuclei and deduces, from Pataraia's fixed-point theorem, an induction principle for joins of nuclei. The scheme (JL) is a formula-level internalization, inside the Heyting algebra, of the common-fixed-point description of the join; what is added here is the deductive system built on it and the study of its expressive power. Nuclei of the form $\bigcirc_C$ on the up-sets of a finite poset are the finite case of the representation of nuclear Heyting algebras and nuclear implicative semilattices by posets with a distinguished subset [2, 1]. For a finite category the Grothendieck topologies correspond to idempotent two-sided ideals (Kelly–Lawvere [11], Marquès [13]), and the join topology to the largest idempotent ideal in the intersection, reached by $2m$ alternating products [15]; part (b) is the poset case of this dictionary.

What is new. The deductive system **LJ**, its algebraic completeness, finite model property and decidability, the alternation bound of Theorem 4.1(ii) with its sharpness, and the inexpressibility Theorem 5.1: the join of two geometric modalities is a genuinely new connective, uniquely determined by the two but not definable from them, in the same way as common knowledge is new relative to the individual knowledge operators (Corollary 5.3).

2. Nuclei and their joins

Let H be a Heyting algebra and $\mathrm{N}(H)$ the set of its nuclei, ordered pointwise. For $j \in \mathrm{N}(H)$ write $\mathrm{Fix}(j) = \{a : ja = a\}$; it is closed under meets and under $b \to (-)$ for every $b \in H$, i.e. $b \to a \in \mathrm{Fix}(j)$ whenever $a \in \mathrm{Fix}(j)$, since $j(b \to a) \le jb \to ja = jb \to a \le b \to a$. Conversely a subset $S \subseteq H$ closed under finite meets and under $b \to (-)$ is the fixed set of at most one nucleus, $j_S(a) = \bigwedge\{s \in S : a \le s\}$ when this meet exists for all a (always if H is complete or finite).

Lemma 2.1. *Let $j_1, j_2 \in \mathrm{N}(H)$. If the meet $ja := \bigwedge\{s \in \mathrm{Fix}(j_1) \cap \mathrm{Fix}(j_2) : a \le s\}$ exists for every $a \in H$, then j is a nucleus, $\mathrm{Fix}(j) = \mathrm{Fix}(j_1) \cap \mathrm{Fix}(j_2)$, and j is the least nucleus above j_1 and j_2 in $\mathrm{N}(H)$. If H is complete the meet always exists, and $j = j_1 \vee j_2$ in the complete lattice $\mathrm{N}(H)$.*

Proof. $S = \mathrm{Fix}(j_1) \cap \mathrm{Fix}(j_2)$ is closed under meets and under $b \to (-)$, so $j = j_S$ is a nucleus with fixed set S (the standard argument: j is inflationary, monotone and idempotent; $j(a \wedge b) \ge ja \wedge jb$ because $a \to j(a \wedge b) \in S$ contains b, hence jb, so $a \wedge jb \le j(a \wedge b)$, then $jb \to j(a \wedge b) \in S$ contains a, hence ja). Any nucleus $k \ge j_1, j_2$ has $\mathrm{Fix}(k) \subseteq S$, so $ja \le ka$; and $j \ge j_i$ since $\mathrm{Fix}(j) \subseteq \mathrm{Fix}(j_i)$. For complete H see [10, II.2.3]. □

Lemma 2.2. *In a complete Heyting algebra, $(j_1 \vee j_2)(a)$ is the least fixed point of $X \mapsto a \vee j_1X \vee j_2X$, and it is $\bigvee_\alpha (j_1j_2)^\alpha(a)$ over the transfinite iterates (Macnab [12]); on a finite algebra it is $(j_1j_2)^n(a)$ for some n.*

Proof. A fixed point X of the operator is above a and fixed by j_1 and j_2, so $X \geq (j_1 \vee j_2)(a)$; and $(j_1 \vee j_2)(a)$ is a fixed point. The iterates $(j_1 j_2)^\alpha(a)$ form an increasing chain of elements below $(j_1 \vee j_2)(a)$ which stabilizes at a common fixed point. □

3. The logic **LJ**

Formulas are built from propositional variables with $\wedge, \vee, \to, \bot, \top$ and three unary connectives $\bigcirc_1, \bigcirc_2, \bigcirc$. For a unary connective $\Box$ the *lax axioms* are

$$\varphi \to \Box\varphi, \qquad \Box\Box\varphi \to \Box\varphi, \qquad \Box\varphi \wedge \Box\psi \to \Box(\varphi \wedge \psi),$$

with the rule: from $\varphi \to \psi$ infer $\Box\varphi \to \Box\psi$. **PLL** is intuitionistic propositional logic with one lax connective [6], $\mathbf{PLL}_2$ has two, $\bigcirc_1$ and $\bigcirc_2$.

Definition 3.1. **LJ** is intuitionistic propositional logic with the lax axioms and rules for $\bigcirc_1$, $\bigcirc_2$ and $\bigcirc$, the axioms $\bigcirc_1\varphi \to \bigcirc\varphi$ and $\bigcirc_2\varphi \to \bigcirc\varphi$, and the induction scheme (JL).

An **LJ**-*algebra* is a Heyting algebra H with nuclei j_1, j_2, j such that $j \geq j_1$, $j \geq j_2$ and, for all $a, b \in H$,

$$(j_1 b \to b) \wedge (j_2 b \to b) \wedge (a \to b) \leq ja \to b. \tag{JL$'$}$$

Since (JL$'$) is an identity, **LJ**-algebras form a variety. **LJ** is algebraizable in the sense of Blok and Pigozzi [3] with this variety as equivalent algebraic semantics, with defining equation $\varphi \approx \top$ and equivalence formulas $\{\varphi \to \psi,\ \psi \to \varphi\}$: this is the algebraizing pair of intuitionistic propositional logic, and it remains one for an axiomatic extension by unary connectives $\Box$ that are *congruential*, i.e. for which $\varphi \leftrightarrow \psi$ yields $\Box\varphi \leftrightarrow \Box\psi$; the monotonicity rule gives this for $\bigcirc_1, \bigcirc_2, \bigcirc$. The axioms translate into the identities defining **LJ**-algebras, (JL) into (JL$'$).

Proposition 3.2. *In an* **LJ**-*algebra, ja is the least element of* $\mathrm{Fix}(j_1) \cap \mathrm{Fix}(j_2)$ *above a; hence j is the least nucleus above j_1 and j_2, and* $\mathrm{Fix}(j) = \mathrm{Fix}(j_1) \cap \mathrm{Fix}(j_2)$. *Conversely, if H is a Heyting algebra with nuclei j_1, j_2 for which the meets of Lemma 2.1 exist (in particular if H is complete), then $(H, j_1, j_2, j_1 \vee j_2)$ is an* **LJ**-*algebra.*

Proof. Let $b \geq a$ be fixed by j_1 and j_2. The left side of (JL$'$) is $\top$, so $ja \leq b$. As $j \geq j_1, j_2$, $\mathrm{Fix}(j) \subseteq \mathrm{Fix}(j_1) \cap \mathrm{Fix}(j_2)$, and ja is fixed by j_1, j_2; so ja is the least common fixed point above a and the rest follows as in Lemma 2.1. For the converse, let $j = j_1 \vee j_2$ and $c \leq (j_1 b \to b) \wedge (j_2 b \to b) \wedge (a \to b)$. Then $c \wedge j_i b \leq b$, so $j_i b \leq c \to b$ and $j_i(c \to b) \leq j_i c \to j_i b \leq c \to j_i b \leq c \to (c \to b) = c \to b$: the element $c \to b$ is fixed by j_1 and j_2, and it is above a because $c \leq a \to b$. Hence $ja \leq c \to b$, i.e. $c \leq ja \to b$. □

Theorem 3.3. $\mathbf{LJ} \vdash \varphi$ *if and only if φ is valid in every* **LJ**-*algebra. Every complete Heyting algebra with two nuclei j_1, j_2, with $\bigcirc$ interpreted as $j_1 \vee j_2$, is an* **LJ**-*algebra; in particular every Grothendieck topos equipped with two*

Lawvere–Tierney topologies and their join validates **LJ** *on its subterminal objects, where the topologies act as nuclei and the join of topologies as the join of nuclei.*

Proof. Soundness for **LJ**-algebras is Proposition 3.2 together with the lax axioms, and completeness is the Lindenbaum–Tarski algebra, which is an **LJ**-algebra because the axioms are identities in it. Complete Heyting algebras with two nuclei are **LJ**-algebras by Proposition 3.2. For the topos statement, the subterminal objects of a Grothendieck topos form a complete Heyting algebra, a Lawvere–Tierney topology restricts to a nucleus on it, and the join of two topologies restricts to the join of the two nuclei, since the restriction $\Omega \to \mathrm{Sub}(1)$ preserves the lattice structure of local operators (both are computed by the same transfinite iteration of the composite, Lemma 2.2). □

Remark 3.4. Theorem 3.3 is completeness for a variety; completeness for the intended complete algebras, where $\bigcirc$ is the join in the lattice of nuclei, follows from the finite model property (Theorem 6.4), since finite algebras are complete. The direct route, embedding an **LJ**-algebra into a completion so that the join of the extended nuclei restricts to j, is not needed.

4. Finite posets and finite sites

Let P be a finite poset and $H = \mathrm{Up}(P)$ its Heyting algebra of up-sets. For $C \subseteq P$ put

$$\textstyle\bigcirc_C U = \{x \in P : {\uparrow}x \cap C \subseteq U\},$$

"U holds at every point of C above x".

Theorem 4.1. (i) *$\bigcirc_C$ is a nucleus on $\mathrm{Up}(P)$, $C \mapsto \bigcirc_C$ is an order-reversing bijection from the subsets of P onto $\mathrm{N}(\mathrm{Up}(P))$, and $\bigcirc_C \wedge \bigcirc_{C'} = \bigcirc_{C\cup C'}$, $\bigcirc_C \vee \bigcirc_{C'} = \bigcirc_{C\cap C'}$.*

(ii) *$\bigcirc_C \vee \bigcirc_{C'} = (\bigcirc_C \bigcirc_{C'})^N$ for every N with $2N > |P|$, in particular for $N = |P|$; and for every N there is a poset with $2N$ points on which $N-1$ iterations do not suffice.*

(iii) *For $2N > |P|$ the scheme $(\bigcirc_1\bigcirc_2)^{N+1}\varphi \to (\bigcirc_1\bigcirc_2)^N\varphi$ is valid on $(\mathrm{Up}(P), \bigcirc_C, \bigcirc_{C'})$, and* **LJ** *plus this scheme proves $\bigcirc\varphi \leftrightarrow (\bigcirc_1\bigcirc_2)^N\varphi$.*

Proof. (i) $\bigcirc_C$ preserves intersections and is inflationary and idempotent (if $x \in \bigcirc_C \bigcirc_C U$ and $c \in C$ is above x, then $c \in \bigcirc_C U$ and $c \in C$ above c gives $c \in U$). For the bijection we use the dictionary of [15, §2]: the nuclei on $\mathrm{Up}(P)$ are the Lawvere–Tierney topologies of the presheaf topos on P, which correspond to Grothendieck topologies on P, hence (Kelly–Lawvere, Marquès, [14, Thm. 5.1]) to idempotent two-sided ideals of the category P, and for a finite poset an idempotent ideal is generated by the identities it contains, i.e. by a subset $C \subseteq P$: the ideal $D_C = \{(x, y) : x \le c \le y \text{ for some } c \in C\}$. The nucleus of D_C on sieves (down-sets) is $S \mapsto \{y : D_C(-, y) \subseteq S\} = \{y : {\downarrow}y \cap C \subseteq S\}$, which is $\bigcirc_C$ after

passing from down-sets to up-sets. Meets of nuclei are pointwise, and $\bigcirc_C U \cap \bigcirc_{C'} U = \bigcirc_{C \cup C'} U$. Joins: $\mathrm{Fix}(\bigcirc_C) = \{U : \forall x\, (\uparrow x \cap C \subseteq U \Rightarrow x \in U)\}$, and $\mathrm{Fix}(\bigcirc_C) \cap \mathrm{Fix}(\bigcirc_{C'}) = \mathrm{Fix}(\bigcirc_{C \cap C'})$: the inclusion $\supseteq$ holds because $\uparrow x \cap C \subseteq U$ implies $\uparrow x \cap (C \cap C') \subseteq U$; for $\subseteq$, let U be fixed by both and $x \notin U$ with $\uparrow x \cap C \cap C' \subseteq U$; choose x maximal with these properties; as U is $\bigcirc_C$-fixed there is $c \in C$ above x with $c \notin U$, and $c \notin C'$ (else $c \in U$); as $c \notin U$ is $\bigcirc_{C'}$-fixed-violating, there is $c' \in C'$ above c with $c' \notin U$, $c' \notin C$; then $c' > x$ has the same properties, contradicting maximality (note $\uparrow c' \cap C \cap C' \subseteq \uparrow x \cap C \cap C' \subseteq U$). By Lemma 2.1 the join is $\bigcirc_{C \cap C'}$.

(ii) By induction, $x \in (\bigcirc_C \bigcirc_{C'})^n U$ iff every alternating chain $x \leq c_1 \leq c_1' \leq c_2 \leq \cdots \leq c_n'$ with $c_i \in C$, $c_i' \in C'$ ends in U. If $x \notin \bigcirc_{C \cap C'} U$, there is $z \in C \cap C'$ above x with $z \notin U$, and the constant chain $c_i = c_i' = z$ shows $x \notin (\bigcirc_C \bigcirc_{C'})^n U$ for all n. Conversely let $x \in \bigcirc_{C \cap C'} U$ and $2N > |P|$, and let $x \leq c_1 \leq c_1' \leq \cdots \leq c_N'$ be an alternating chain. Its $2N > |P|$ entries cannot all be distinct, and since the chain is monotone two adjacent entries coincide: $c_i = c_i'$ or $c_i' = c_{i+1}$, a point of $C \cap C'$ above x, hence in U; the chain stays above it and ends in U. For the second claim take P the chain $1 < 2 < \cdots < 2N$, C the odd points together with $2N$, C' the even points, and $U = \{2N\}$: then $C \cap C' = \{2N\}$ and $1 \in \bigcirc_{C \cap C'} U$, while the chain $1 \leq 1 \leq 2 \leq 3 \leq 4 \leq \cdots \leq 2N - 2$ of $2(N-1)$ entries ends outside U, so $1 \notin (\bigcirc_C \bigcirc_{C'})^{N-1} U$.

(iii) The first claim is (ii). In **LJ** plus the scheme, $\theta := (\bigcirc_1 \bigcirc_2)^N \varphi$ satisfies $\varphi \to \theta$, $\bigcirc_1 \theta \to \theta$ (as $\bigcirc_1 \theta = \bigcirc_1 (\bigcirc_1 \bigcirc_2)^N \varphi$ is provably equivalent to $(\bigcirc_1 \bigcirc_2)^N \varphi$ by idempotence of $\bigcirc_1$) and $\bigcirc_2 \theta \to \theta$ (as $\bigcirc_2 (\bigcirc_1 \bigcirc_2)^N \varphi \to \bigcirc_1 \bigcirc_2 (\bigcirc_1 \bigcirc_2)^N \varphi = (\bigcirc_1 \bigcirc_2)^{N+1} \varphi \to \theta$ by the scheme), so (JL) gives $\bigcirc \varphi \to \theta$; and $\theta \to \bigcirc \varphi$ since $\bigcirc_1, \bigcirc_2 \leq \bigcirc$ and $\bigcirc$ is idempotent. □

Remark 4.2. For a finite category S in place of a poset, the nuclei on the subterminal objects of $\mathrm{Presh}(S)$ are again the Grothendieck topologies, i.e. the idempotent ideals D, with $\bigcirc_D$ given by $D(-, c) \subseteq S$; the join of $\bigcirc_D$ and $\bigcirc_E$ is $\bigcirc_{I'}$ for the largest idempotent ideal $I' \subseteq D \cap E$, and it is reached by $2m$ alternating composites, where m is the length of the chain $DE \supseteq DEDE \supseteq \cdots$ [15, §2]. The sameness-lattice theorem of [15], $W_D \vee W_E = W_{I'}$, is the statement that the maps of presheaves f with $\bigcirc_{I'} [\![f \text{ is an isomorphism}]\!] = \top$ are exactly the two-out-of-three combinations of maps g with $\bigcirc_D [\![g \text{ is an isomorphism}]\!] = \top$ or $\bigcirc_E [\![g \text{ is an isomorphism}]\!] = \top$; it is a statement about maps and does not reduce to the propositional logic **LJ**.

5. Inexpressibility

Theorem 5.1. *There is no formula χ of $\mathbf{PLL}_2$ (in the connectives $\wedge, \vee, \to, \bot, \top, \bigcirc_1, \bigcirc_2$ and propositional variables) such that $\bigcirc \bot \leftrightarrow \chi$ is valid in all **LJ**-algebras, nor in all complete chains with two nuclei. Consequently no $\chi(p)$ defines $\bigcirc p$, and for every n the formula $\bigcirc \bot \leftrightarrow (\bigcirc_1 \bigcirc_2)^n \bot$ fails in some complete chain.*

Proof. Let H be the chain $\omega+2 = \{0 < 1 < 2 < \cdots < \omega < \omega+1\}$, a complete Heyting algebra with $a \to b = \top$ if $a \le b$ and $a \to b = b$ otherwise. On a chain every inflationary idempotent monotone map is a nucleus. Let

$$j_1(2k) = 2k{+}1, \quad j_1(2k{+}1) = 2k{+}1, \qquad j_2(2k{+}1) = 2k{+}2, \quad j_2(2k) = 2k,$$

and $j_1(\omega) = j_2(\omega) = \omega$, $j_1(\omega{+}1) = j_2(\omega{+}1) = \omega{+}1$. Then $\mathrm{Fix}(j_1) = \{\text{odd}\} \cup \{\omega, \omega+1\}$, $\mathrm{Fix}(j_2) = \{\text{even}\} \cup \{\omega, \omega+1\}$, so by Lemma 2.1 $(j_1 \vee j_2)(a) = \omega$ for all $a \le \omega$, in particular $(j_1 \vee j_2)(\bot) = \omega$. The set $A = \mathbb{N} \cup \{\omega+1\}$ contains $\bot = 0$ and $\top = \omega+1$, is closed under j_1, j_2, under $\wedge, \vee$, and under $\to$ (as $a \to b \in \{\top, b\}$). If χ had the stated property, then, whatever values in H are assigned to its variables — assign $\bot$ to all of them — the value of χ would be ω; but the value of a formula all of whose variables take values in A lies in A, and $\omega \notin A$. The second statement follows by substituting $\bot$ for p; the third because $(j_1 j_2)^n(0) = 2n \neq \omega$. □

Remark 5.2. The join is implicitly definable: in any Heyting algebra with nuclei j_1, j_2 there is at most one nucleus j satisfying the **LJ**-axioms (Proposition 3.2). Theorem 5.1 is thus a failure of explicit definability for a connective, of the same kind as the inexpressibility of common knowledge by finite iterations of "everybody knows" [9]: there $C\varphi = \bigwedge_n E^n\varphi$ is a greatest fixed point of a deflationary operator and the induction axiom $C(\varphi \to E\varphi) \to (\varphi \to C\varphi)$ expresses it; here $\bigcirc\varphi = \bigvee_n (\bigcirc_1\bigcirc_2)^n\varphi$ (on complete algebras, with transfinite iteration in general) is a least fixed point of an inflationary operator and (JL) expresses it. In the chain of Theorem 5.1 the join needs exactly ω iterations.

Corollary 5.3 (uniqueness without definability)**.** *Let* $\mathbf{LJ}^{\bigcirc,\bigcirc'}$ *be* **LJ** *with a second copy* $\bigcirc'$ *of the join connective, satisfying the same axioms relative to* $\bigcirc_1, \bigcirc_2$*. Then* $\mathbf{LJ}^{\bigcirc,\bigcirc'} \vdash \bigcirc\varphi \leftrightarrow \bigcirc'\varphi$ *for all* φ*, while no formula* $\chi(p)$ *of* $\mathbf{PLL}_2$ *satisfies* $\mathbf{LJ} \vdash \bigcirc p \leftrightarrow \chi(p)$*. Thus the join connective is implicitly but not explicitly definable over* $\mathbf{PLL}_2$*, in the sense in which a connective is implicitly defined by a set of axioms when any two connectives satisfying them are provably equivalent; the analogue for connectives of the Beth definability property thus fails for this extension.*

Proof. In an algebra of the extended logic both j and j' send a to the least common fixed point of j_1, j_2 above a (Proposition 3.2), so $j = j'$ and the equivalence is valid, hence provable by algebraic completeness. The second claim is Theorem 5.1. □

Remark 5.4. Everything extends to r lax modalities $\bigcirc_1, \ldots, \bigcirc_r$ and their join $\bigcirc$: the axioms are $\bigcirc_i\varphi \to \bigcirc\varphi$ and $(\bigcirc_1\psi \to \psi) \to \cdots \to (\bigcirc_r\psi \to \psi) \to (\varphi \to \psi) \to (\bigcirc\varphi \to \psi)$, the algebras are those in which $\bigcirc a$ is the least common fixed point of $\bigcirc_1, \ldots, \bigcirc_r$ above a, on a finite poset the join of $\bigcirc_{C_1}, \ldots, \bigcirc_{C_r}$ is $\bigcirc_{C_1 \cap \cdots \cap C_r}$, and the chain of Theorem 5.1 with $r-2$ identity nuclei added shows inexpressibility.

6. The finite model property

Let L be a Heyting algebra and X the set of its prime filters, ordered by inclusion and with the Priestley topology generated by the sets $\hat{V} = \{x : V \in x\}$ and their complements, $V \in L$; X is an Esakia space, $V \mapsto \hat{V}$ is an isomorphism of L onto the clopen up-sets of X, and $\widehat{V \to W} = \{x : \forall y \supseteq x\,(V \in y \Rightarrow W \in y)\}$ [4, Ch. 11].

Lemma 6.1. *Every nonempty closed subset F of X has a maximal element above each of its points.*

Proof. Let $x \in F$. By Zorn's lemma it suffices that every chain $\mathcal{C}$ in $F \cap {\uparrow} x$ has an upper bound in $F \cap {\uparrow} x$. For $y \in X$ the set ${\uparrow} y = \bigcap_{V \in y} \hat{V}$ is closed, and the family $\{F \cap {\uparrow} y : y \in \mathcal{C}\}$ of closed sets has the finite intersection property (it is a chain of nonempty sets), so by compactness its intersection is nonempty; any of its points is an upper bound. □

Lemma 6.2 (nuclei as relativized necessities)**.** *Let j be a nucleus on L and $S_j = \{z \in X : jV \in z \Rightarrow V \in z \text{ for all } V \in L\}$, the* j-closed *prime filters. Then for every $V \in L$,*

$$\widehat{jV} = \{x \in X : \forall z \supseteq x\,(z \in S_j \Rightarrow V \in z)\} = \textstyle\bigcirc_{S_j} \hat{V}.$$

Proof. If $jV \in x \subseteq z \in S_j$ then $jV \in z$, so $V \in z$. Conversely let $jV \notin x$ and let $z \supseteq x$ be a filter maximal with $jV \notin z$. Then z is prime, $V \notin z$, and z is j-closed: if $jW \in z$ and $W \notin z$, the filter generated by z and W contains jV, so $w \wedge W \leq jV$ for some $w \in z$, whence $jw \wedge jW \leq jV$ and $jV \in z$, a contradiction. □

For an **LJ**-algebra (L, j_1, j_2, j) write S_1, S_2, S for the sets of j_1-, j_2-, j-closed prime filters. Always $S \subseteq S_1 \cap S_2$, since $j_i V \leq jV$; the inclusion can be strict: on the chain $\omega + 2$ of Theorem 5.1 the filter ${\uparrow}\omega$ is j_1- and j_2-closed, but $j(0) = \omega \in {\uparrow}\omega$ while $0 \notin {\uparrow}\omega$. The following lemma, which is where the induction scheme does its work, says that a point of $S_1 \cap S_2$ which is isolated from above by an element of L lies in S.

Lemma 6.3 (isolated points are closed)**.** *Let $m \in S_1 \cap S_2$ and suppose there is $G \in L$ with $G \notin m$ such that $G \in y$ for every prime filter $y \supsetneq m$. Then $m \in S$.*

Proof. Let $U \notin m$; we show $jU \notin m$. Put $b = U \vee G$; then $b \notin m$ since m is prime. We claim $j_i b \to b \in m$ for $i = 1, 2$ and $U \to b \in m$. The last is clear as $U \leq b$. For the first, let $y \supseteq m$ with $j_i b \in y$: if $y \supsetneq m$ then $G \in y$, so $b \in y$; if $y = m$ then $b \in m$ because m is j_i-closed. Hence $m \in \widehat{j_i b \to b}$. By (JL′), $(j_1 b \to b) \wedge (j_2 b \to b) \wedge (U \to b) \leq jU \to b$, so $jU \to b \in m$; if $jU \in m$ then $b \in m$, a contradiction. □

Theorem 6.4. *If* **LJ** $\nvdash \varphi$*, then φ is refuted on a finite poset W with two subsets C_1, C_2, with $\bigcirc_1, \bigcirc_2, \bigcirc$ interpreted as $\bigcirc_{C_1}, \bigcirc_{C_2}, \bigcirc_{C_1 \cap C_2}$ on* $\mathrm{Up}(W)$*. Consequently* **LJ** *is decidable, and it is complete for the finite, and*

for the complete, Heyting algebras with two nuclei and their join, and for Grothendieck toposes with two Lawvere–Tierney topologies and their join.

Proof. Let φ be refuted in an **LJ**-algebra L under a valuation, and let Cl be the finite set of subformulas of φ. For a prime filter x put $T(x) = \{\gamma \in \mathrm{Cl} : [\![\gamma]\!] \in x\}$; the set $Q(x) = \{y \supseteq x : T(y) = T(x)\}$ is closed, and T is monotone. Call a prime filter m *settled* if it is a maximal element of $Q(y)$ for some y; then every $z \supsetneq m$ has $T(z) \supsetneq T(m)$, so with $G_m = \bigvee\{[\![\gamma]\!] : \gamma \in \mathrm{Cl} \setminus T(m)\}$ (which is $\bot$ if $T(m) = \mathrm{Cl}$) we have $G_m \notin m$ and $G_m \in z$ for all $z \supsetneq m$; by Lemma 6.3, *a settled point of $S_1 \cap S_2$ lies in S.*

Settled points exist above every point by Lemma 6.1, $Q(x)$ being closed. Choose x_0 with $[\![\varphi]\!] \notin x_0$ and a settled $m_0 \in Q(x_0)$. We select a finite set W of settled prime filters by recursion: having selected m, for each formula of Cl not in $T(m)$ we select a witness as follows, each with a strictly larger T.

- $\alpha \to \beta \notin T(m)$ with $\alpha \notin T(m)$: there is $y \supseteq m$ with $\alpha \in T(y)$, $\beta \notin T(y)$; select a settled $m' \in Q(y)$. (If $\alpha \in T(m)$ then $\beta \notin T(m)$ and m itself is the witness.)
- $\bigcirc_i\beta \notin T(m)$ with $m \notin S_i$ $(i = 1, 2)$: by Lemma 6.2 there is $y \supseteq m$ in S_i with $\beta \notin T(y)$; if $T(y) = T(m)$ then $y = m$, since every $z \supsetneq m$ has $T(z) \supsetneq T(m)$, contradicting $m \notin S_i$; hence $T(y) \supsetneq T(m)$. Let m' be a settled point of $Q(y)$; if $m' \notin S_i$, repeat from m' (still $\bigcirc_i\beta \notin T(m')$, since $y \in S_i$ and $\beta \notin T(y) = T(m')$). As T strictly increases, this stops at a settled point $m'' \in S_i$ with $T(m'') \supsetneq T(m)$ and $\beta \notin T(m'')$; select m'' (the intermediate points are not selected). (If $m \in S_i$, m is its own witness, as $\beta \notin T(m)$.)
- $\bigcirc\alpha \notin T(m)$ with $m \notin S$: the same with S in place of S_i, using Lemma 6.2 for j.

Each selected point has at most $|\mathrm{Cl}|$ selected witnesses, each with a strictly larger T, so the recursion has depth at most $|\mathrm{Cl}|$ and W is finite. Order W by inclusion and put $C_i = W \cap S_i$; then $W \cap S \subseteq C_1 \cap C_2$, and $C_1 \cap C_2 \subseteq S$ because selected points are settled. Interpret the variables by $V(p) = \{m \in W : p \in T(m)\}$. We claim that for all $\gamma \in \mathrm{Cl}$ and $m \in W$: $m \Vdash \gamma$ iff $\gamma \in T(m)$. Variables, $\wedge$ and $\vee$ are clear, $T(m)$ coming from a prime filter. For $\alpha \to \beta$: if $\alpha \to \beta \in T(m)$ and $n \supseteq m$ in W with $n \Vdash \alpha$, then $\alpha, \alpha \to \beta \in T(n)$, so $\beta \in T(n)$; the converse is the selected witness. For $\bigcirc_i\beta$: if $\bigcirc_i\beta \in T(m)$ and $n \in C_i$, $n \supseteq m$, then $\bigcirc_i\beta \in T(n)$ and n is j_i-closed, so $\beta \in T(n)$; the converse is the selected witness, which lies in C_i. For $\bigcirc\alpha$: if $\bigcirc\alpha \in T(m)$ and $n \in C_1 \cap C_2$, $n \supseteq m$, then $n \in S$, so $\alpha \in T(n)$; the converse is the selected witness, which lies in $W \cap S \subseteq C_1 \cap C_2$. By Theorem 4.1, $(\mathrm{Up}(W), \bigcirc_{C_1}, \bigcirc_{C_2}, \bigcirc_{C_1 \cap C_2})$ is a finite **LJ**-algebra, and it refutes φ at m_0.

Decidability follows since **LJ** is finitely axiomatized and validity on a finite poset with two subsets is decidable; the completeness statements follow

because finite Heyting algebras are complete and are the algebras of subterminal objects of presheaf toposes on finite posets, on which $\bigcirc_C$ is the Grothendieck topology generated by C. □

Corollary 6.5. *If* $\mathbf{LJ} \nvdash \varphi$ *and* n *is the number of subformulas of* φ*, then* φ *is refuted on a poset with at most* $1 + n + n^2 + \cdots + n^n$ *points and two subsets. Hence* $\mathbf{LJ} \vdash \varphi$ *can be decided by checking* φ *on all posets of that size with all pairs of subsets and all valuations.*

Proof. In the proof of Theorem 6.4 each selected point has at most n selected witnesses and the depth of the recursion is at most n, as T increases strictly within the n-element set Cl. □

Remark 6.6. The induction scheme enters only through Lemma 6.3, and there only for the elements $b = U \vee G_m$ with G_m definable from Cl. This is the lax counterpart of the finite canonical model construction for common knowledge [9], where the induction axiom is applied to the characteristic formula of a finite set of worlds; here the characteristic formula is replaced by the element G_m which isolates a settled point from above.

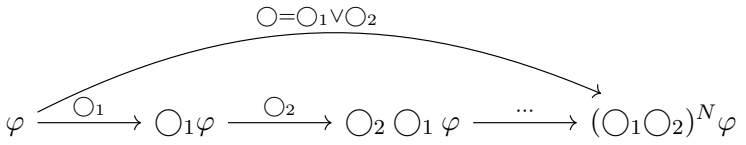


FIGURE 1. The join as the limit of alternating iterates: finite on a finite poset ($N = |P|$, Theorem 4.1), transfinite in general, and not a $\mathbf{PLL}_2$-formula (Theorem 5.1).

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